

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

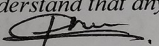
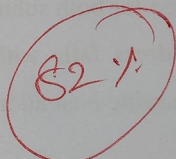
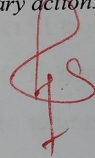
Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Priyav-H (192525420) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately ²	Understands problem with minor gaps ^{1.5}	Partial understanding; misses key elements ¹	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification ²	Selects correct method with minor errors ^{1.5}	Inappropriate or partially correct method ¹	Unable to select method ⁰
Solution Process	25%	Logical systematic steps with correct approach throughout ^{2.5}	Mostly correct steps with minor mistakes ²	Disorganized steps; multiple errors ^{1.5}	No clear process ¹
Accuracy of Calculation	20%	All calculations accurate; correct final answer ²	Minor calculation errors ^{1.5}	Frequent errors; incorrect final answer ¹	No valid calculations ⁰
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance ^{1.5}	Basic interpretation with minor omissions ¹	Limited or unclear interpretation ^{0.5}	No interpretation ⁰

	Problem understanding understanding of scenario (20%)	Method Selection (20%) Identification of key issue (20%)	Solution Process (25%) Application of concept (25%)	Accuracy of calculation (20%) Decision making (20%)	Interpretation (15%) Clarity of Response (15%)
Q1	2	1.5	2	1.5	1 (8)
Q2	1.5	1	1.5	1	1 (6)
Q3	2	2	2.5	2	1.5 (7)
Q4	2	2	2	2	1 (6)
Q5	1.5	1.5	1.5	2	1.5 (8)
				(41)	

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Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING — WITH SOLUTIONS

Question 1: A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

Answer:

Base network: 192.168.20.0/24. Subnets are allocated sequentially, largest block first (Lab A, then Lab B, then Lab C).

Lab A — /26 (Subnet Mask 255.255.255.192, Block size = 64):

- Network Address: 192.168.20.0
- Valid Host Range: 192.168.20.1 – 192.168.20.62
- Broadcast Address: 192.168.20.63
- Usable Hosts: $2^6 - 2 = 62$

Lab B — /27 (Subnet Mask 255.255.255.224, Block size = 32):

- Network Address: 192.168.20.64
- Valid Host Range: 192.168.20.65 – 192.168.20.94
- Broadcast Address: 192.168.20.95
- Usable Hosts: $2^5 - 2 = 30$

Lab C — /28 (Subnet Mask 255.255.255.240, Block size = 16):

- Network Address: 192.168.20.96
- Valid Host Range: 192.168.20.97 – 192.168.20.110
- Broadcast Address: 192.168.20.111
- Usable Hosts: $2^4 - 2 = 14$

Interpretation: The three labs are allocated back-to-back non-overlapping blocks within 192.168.20.0/24 (Lab A: .0–.63, Lab B: .64–.95, Lab C: .96–.111), leaving the remaining addresses (.112–.255) free for future use.

Question 2: A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system.

Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Answer:

Subnet Mask 255.255.255.192 = /26, Block size = 64. The 4th-octet subnet boundaries are: 0, 64, 128, 192.

System 1 — IP 192.168.1.130:

- 130 lies between 128 and 191 → Subnet: 192.168.1.128/26
- Network Address: 192.168.1.128
- Broadcast Address: 192.168.1.191
- Valid Host Range: 192.168.1.129 – 192.168.1.190
- Check: 192.168.1.130 lies within 129–190 → Valid host address in this subnet.

System 2 — IP 192.168.1.190:

- 190 lies between 128 and 191 → Subnet: 192.168.1.128/26
- Network Address: 192.168.1.128
- Broadcast Address: 192.168.1.191
- Valid Host Range: 192.168.1.129 – 192.168.1.190
- Check: 192.168.1.190 lies within 129–190 → Valid host address (last usable host) in this subnet.

Usable Host Addresses per Subnet: $2^6 - 2 = 62$

Interpretation: Both System 1 and System 2 belong to the same subnet, 192.168.1.128/26, since both their addresses fall within the range 192.168.1.129–192.168.1.190. They can therefore communicate directly without routing.

Question 3: An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

Answer:

Base network: 172.16.0.0/16. Subnets are allocated sequentially in decreasing size order: HR, Sales, IT, Admin.

HR — /25 (Mask 255.255.255.128, Block size = 128):

- Network Address: 172.16.0.0

- Valid Host Range: 172.16.0.1 – 172.16.0.126
- Broadcast Address: 172.16.0.127
- Usable Hosts: $2^7 - 2 = 126$

Sales — /26 (Mask 255.255.255.192, Block size = 64):

- Network Address: 172.16.0.128
- Valid Host Range: 172.16.0.129 – 172.16.0.190
- Broadcast Address: 172.16.0.191
- Usable Hosts: $2^6 - 2 = 62$

IT — /27 (Mask 255.255.255.224, Block size = 32):

- Network Address: 172.16.0.192
- Valid Host Range: 172.16.0.193 – 172.16.0.222
- Broadcast Address: 172.16.0.223
- Usable Hosts: $2^5 - 2 = 30$

Admin — /28 (Mask 255.255.255.240, Block size = 16):

- Network Address: 172.16.0.224
- Valid Host Range: 172.16.0.225 – 172.16.0.238
- Broadcast Address: 172.16.0.239
- Usable Hosts: $2^4 - 2 = 14$

Remaining Unused IP Range: 172.16.0.240 – 172.16.255.255 (all addresses in 172.16.0.0/16 beyond 172.16.0.239 remain unallocated and available for future departments/subnets).

Question 4: A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

Answer:

/27 = Mask 255.255.255.224, Block size = 32. Borrowed bits from host portion = 3, so Number of subnets = $2^3 = 8$.

Usable Hosts per Subnet = $2^5 - 2 = 30$

The 8 subnets (4th octet boundaries): 0, 32, 64, 96, 128, 160, 192, 224.

Locating 172.16.10.78:

- 78 lies between 64 and 95 → belongs to subnet 172.16.10.64/27
- Network Address: 172.16.10.64

- Broadcast Address: 172.16.10.95
- Valid Host Range: 172.16.10.65 – 172.16.10.94

Checking 172.16.10.95:

- 172.16.10.95 lies within the same subnet range (64–95) as 172.16.10.78.
- However, 172.16.10.95 is the Broadcast Address of this subnet, not a usable host address — it identifies the subnet as a whole and cannot be assigned to a device.

Interpretation: A /27 mask on 172.16.10.0/24 creates 8 equal subnets of 32 addresses each (30 usable hosts). 172.16.10.78 and 172.16.10.95 both fall in the same subnet (172.16.10.64/27), but only .78 is a valid host address; .95 is reserved for broadcast.

Question 5: A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Answer:

Given Address: 2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1 — Compression Rules: (a) Leading zeros in each group can be dropped, (b) One single contiguous run of all-zero groups can be replaced with '::' (only once).

- Remove leading zeros: 2001:db8:0:0:0:ff00:42:8329
- Replace the contiguous zero groups (3rd–5th groups) with '::' : 2001:db8::ff00:42:8329

Compressed Form: 2001:db8::ff00:42:8329

Step 2 — Expansion back to full notation:

- '::' is replaced with the correct number of all-zero groups to make 8 groups of 4 hex digits total.

Expanded Form: 2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3 — Network Prefix at /64:

- The first 64 bits (first 4 groups) form the network portion; the remaining 64 bits are the interface ID.
- Network Prefix: 2001:0db8:0000:0000::/64 (or in compressed form: 2001:db8::/64)

Step 4 — Total Possible Host Addresses:

- With a /64 prefix, $128 - 64 = 64$ bits are available for host (interface) addressing.
- Total possible addresses = $2^{64} \approx 1.844674 \times 10^{19}$ addresses.

Interpretation: A /64 prefix is the standard IPv6 subnet size, providing an extremely large host address space (2^{64} addresses) per subnet — far more than any physical network could ever need, which simplifies address allocation and supports auto-configuration (SLAAC).

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